

ML/IA

Workshop 01

Thiago Hersan

Workshop 01

- apresentação
- conceitos e terminologia

intro

Interfaces de navegação para acervos de interesse público

Found Works: 263
[List of Works](#)

Instituto Hércules Florence
Museu Histórico Nacional
Coleção Brasileira Itau

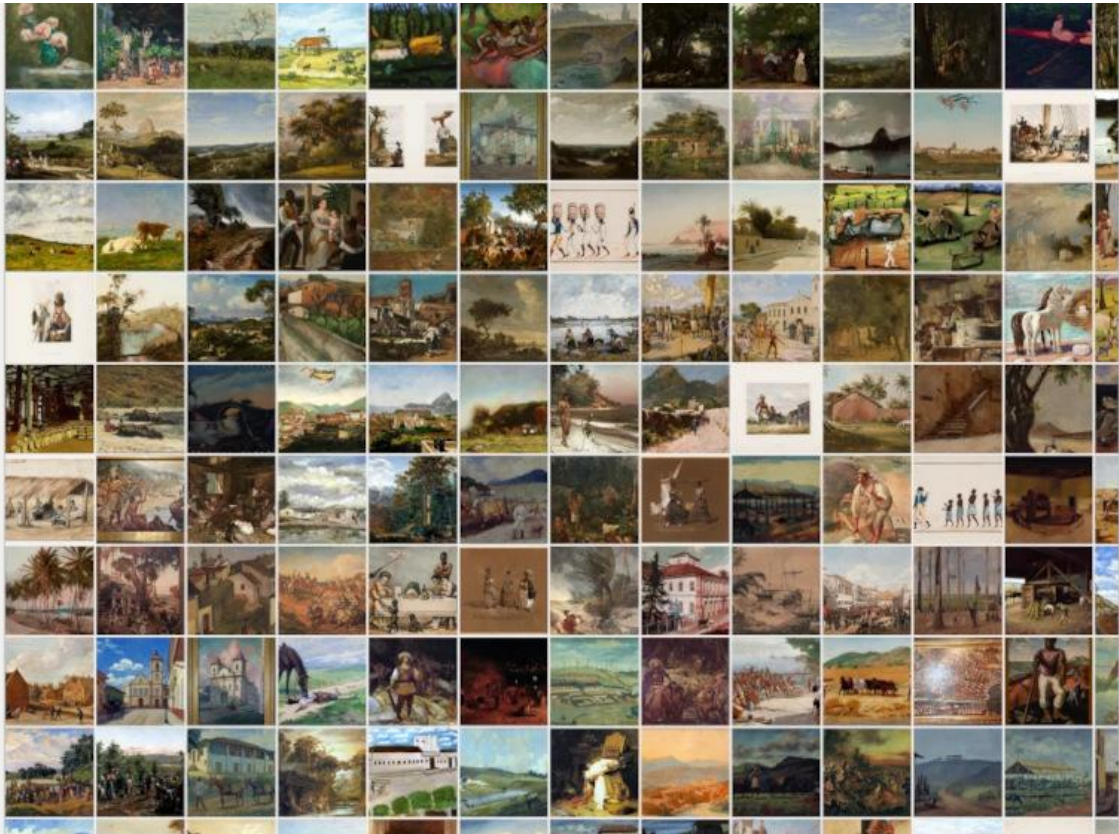
All None

Techniques
Select type of work or technique
drawing painting

Date
Select date range by year of creation
From 1275 To 2021

Fauna and Flora
Select elements of interest
bird dog horse ox bush
flower fruit grass greenery
shrub tree vegetation conifer
palm tree person face hand
back breast gluteous torso

Clustering
Select by groups determined by AI
figure studies religious figures
landscape paintings



Visualizations
How to display the selected content

Timeline
By year of creation

Chromatic Ordering
By dominant colors
Select a color



Organize by color and year

Cluster / Cluster
2D visualizations of clusters

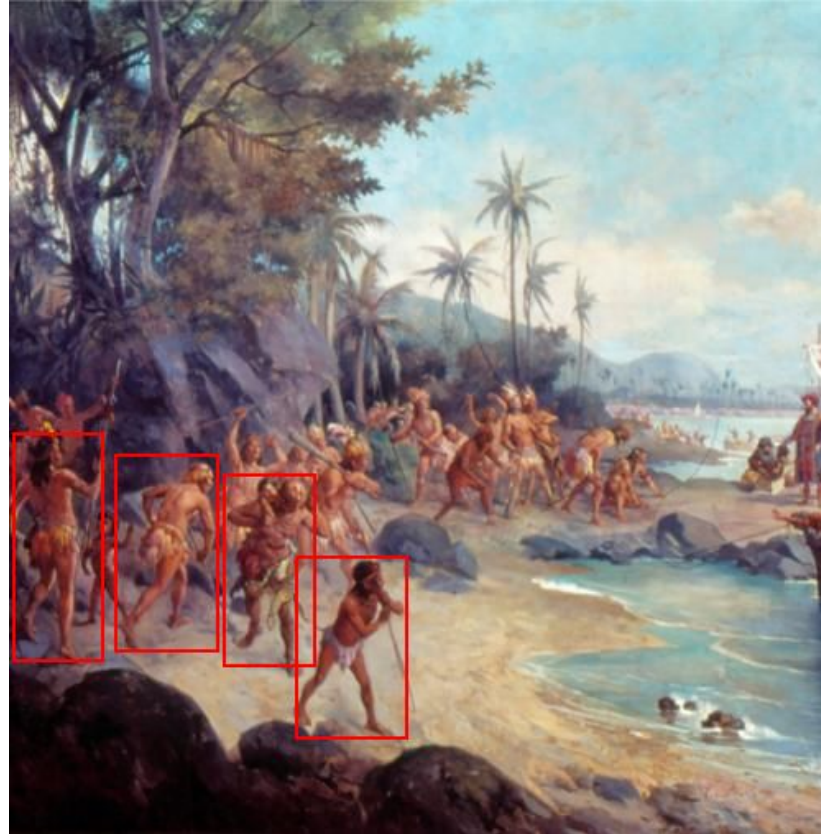
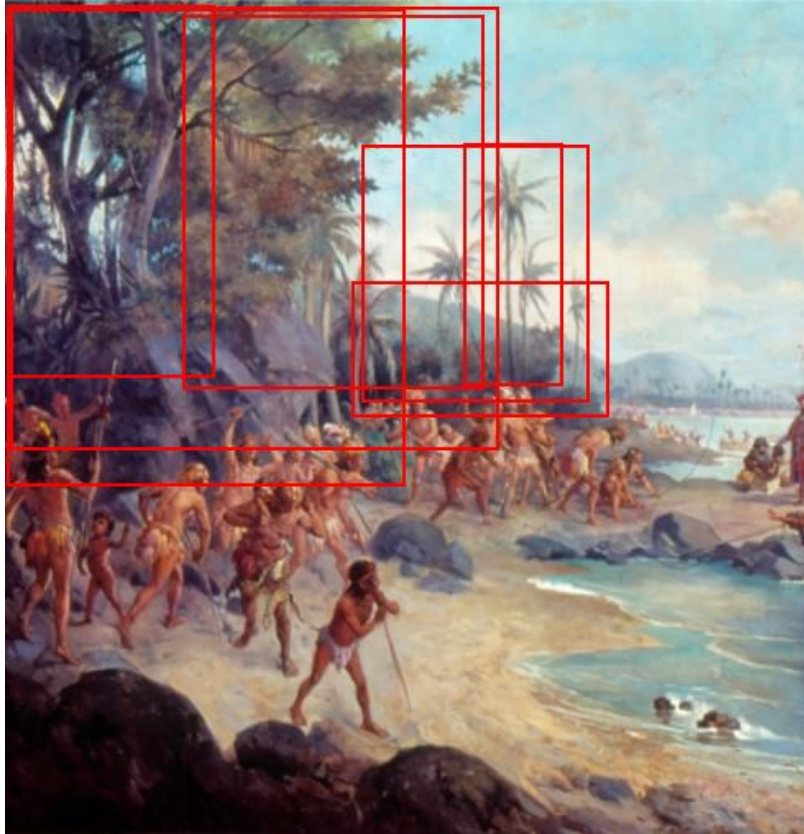
Export

[LINK](#)

Visualização / Organização / Navegação



Modelos de visão computacional (IA/ML)

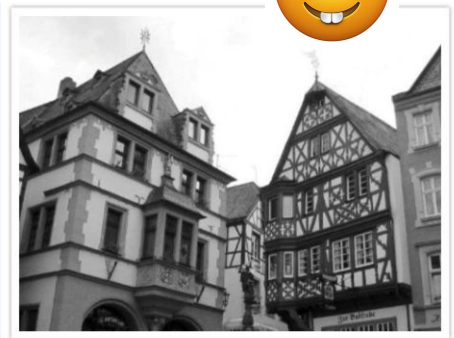


o que é um modelo ?

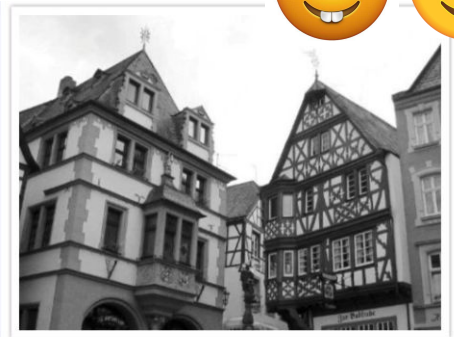
Paradigmas de programação



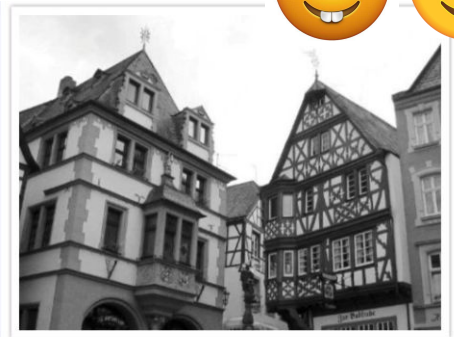
Programação clássica



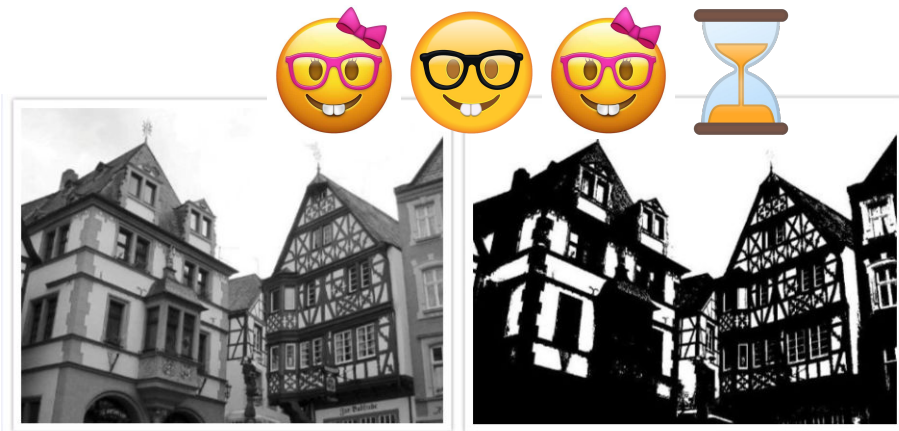
Programação clássica



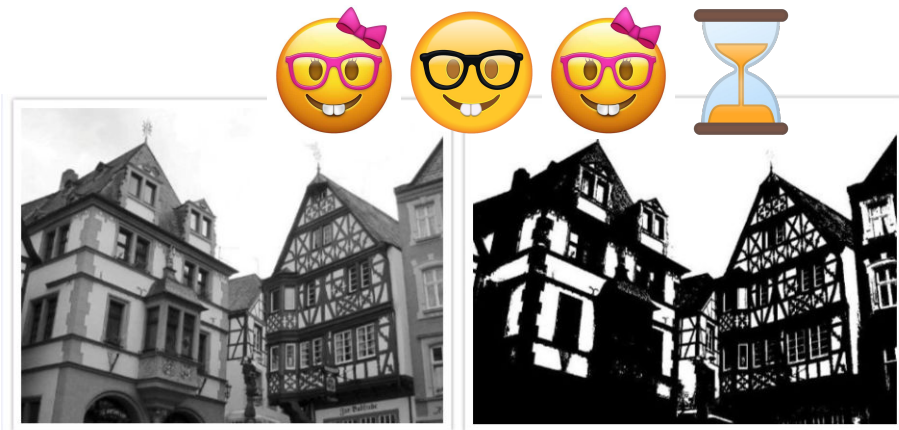
Programação clássica



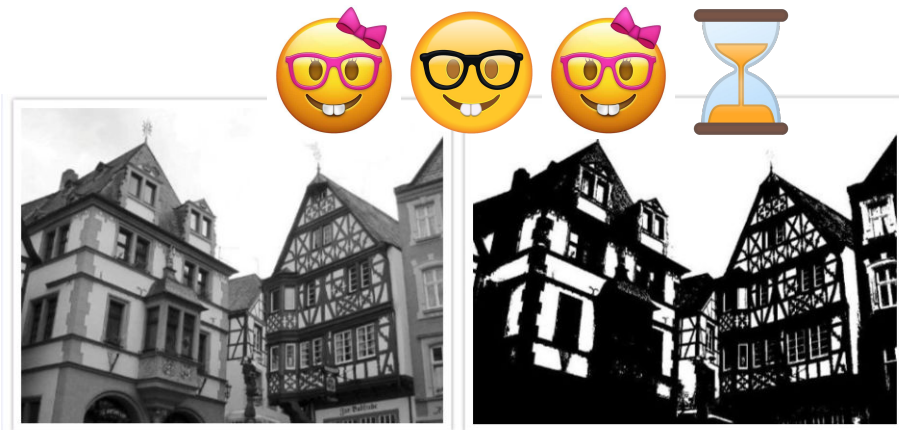
Programação clássica



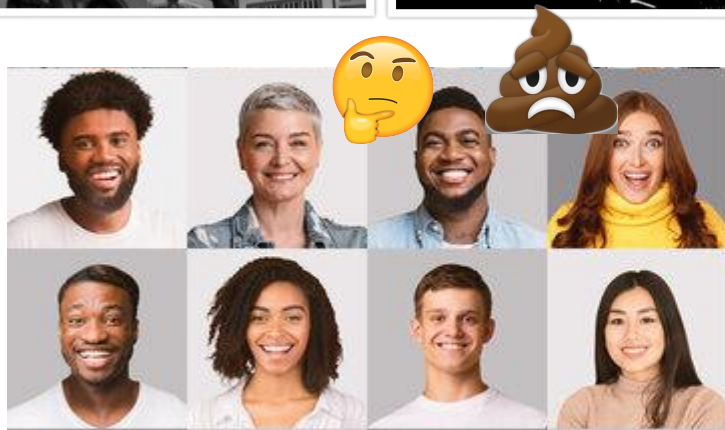
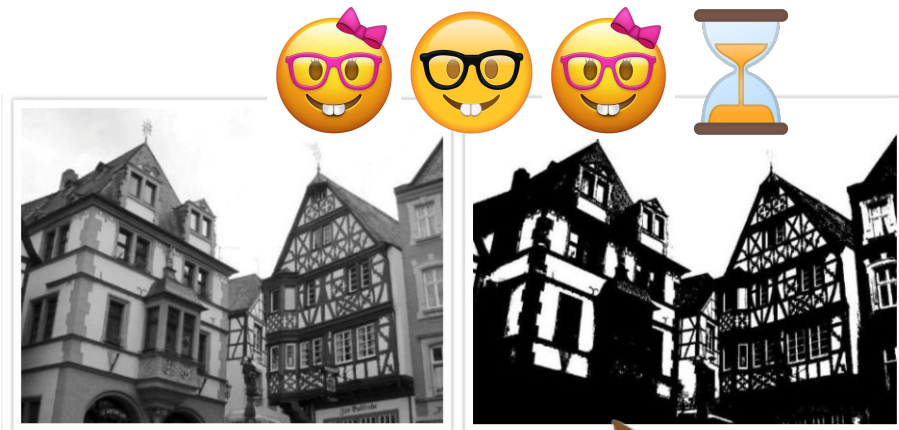
Programação clássica



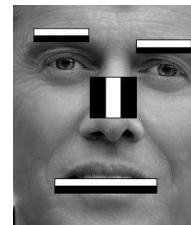
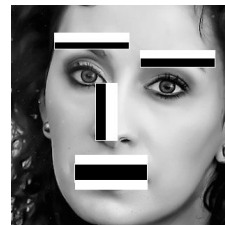
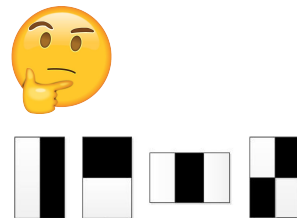
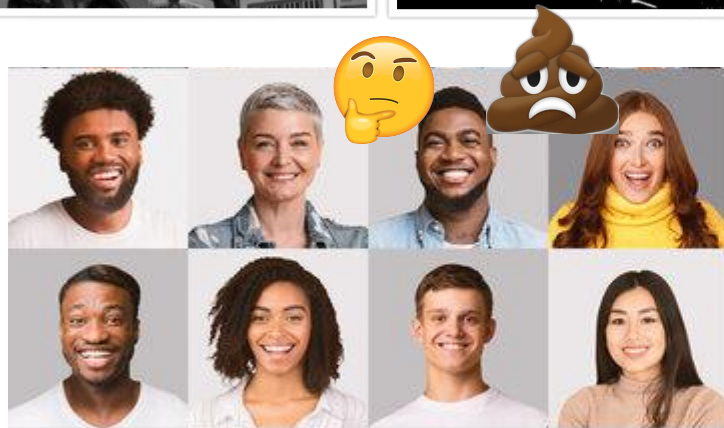
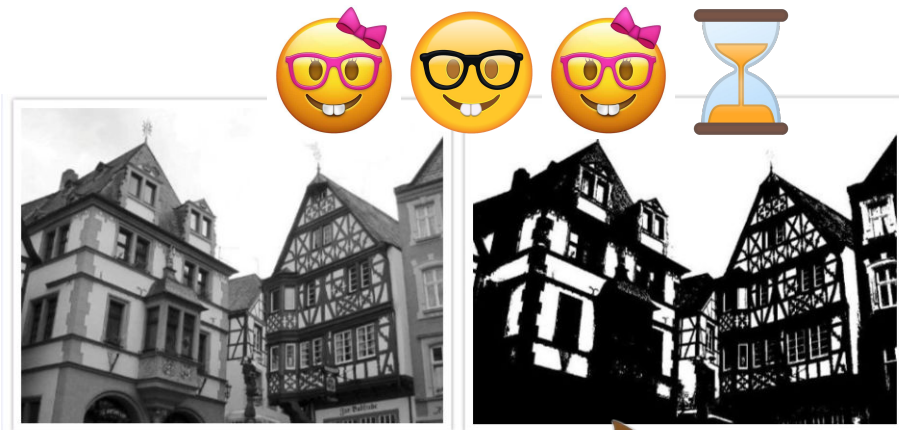
Programação clássica



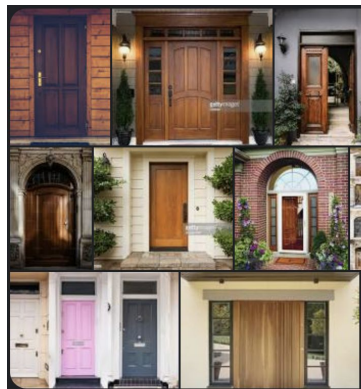
Programação clássica



Programação clássica

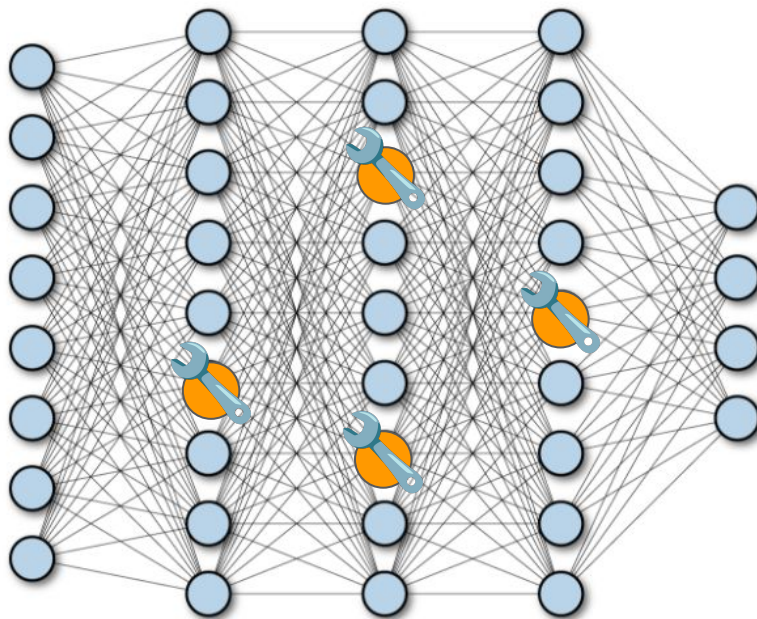


Programação com dados e aprendizado de máquinas



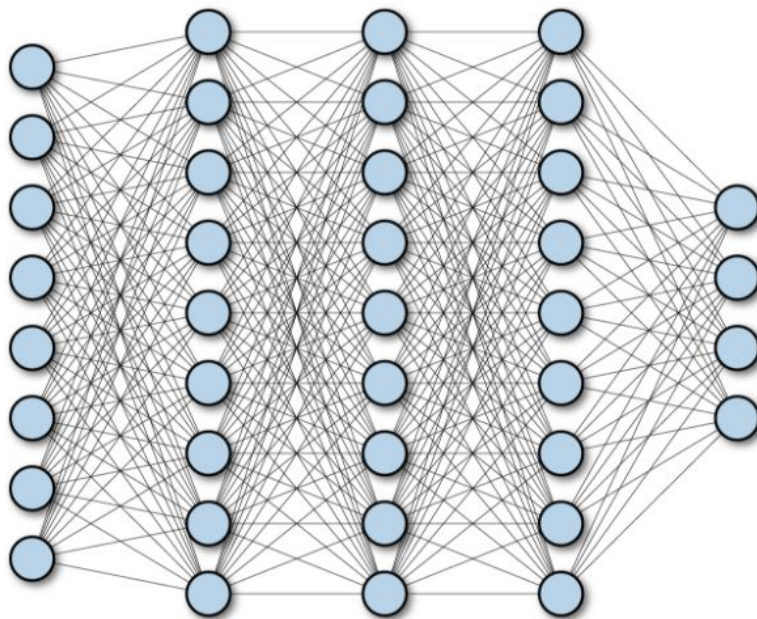
Programação com dados e aprendizado de máquinas

Treinamento



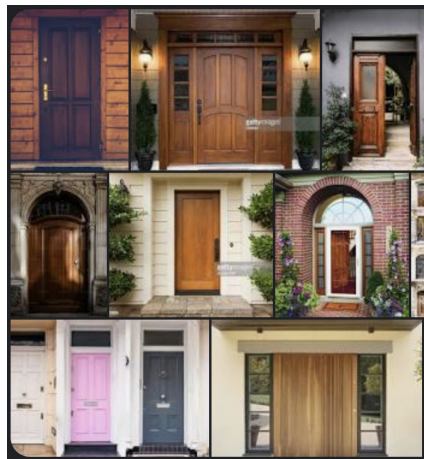
Programação com dados e aprendizado de máquinas

Inferência

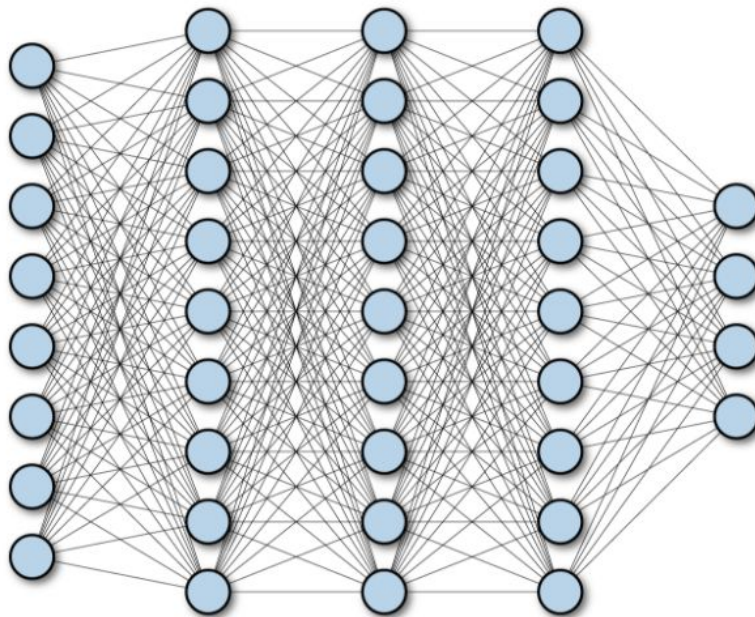


Programação com dados e aprendizado de máquinas

Dados



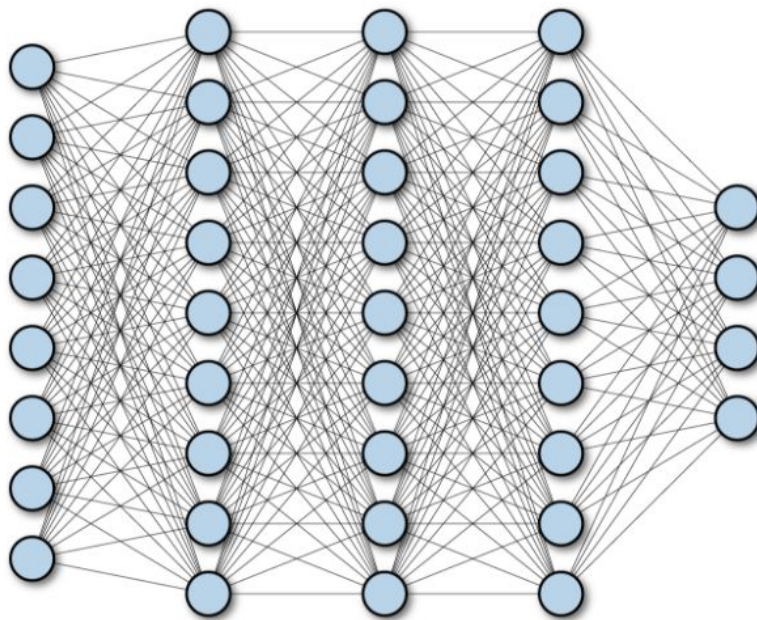
Treinamento



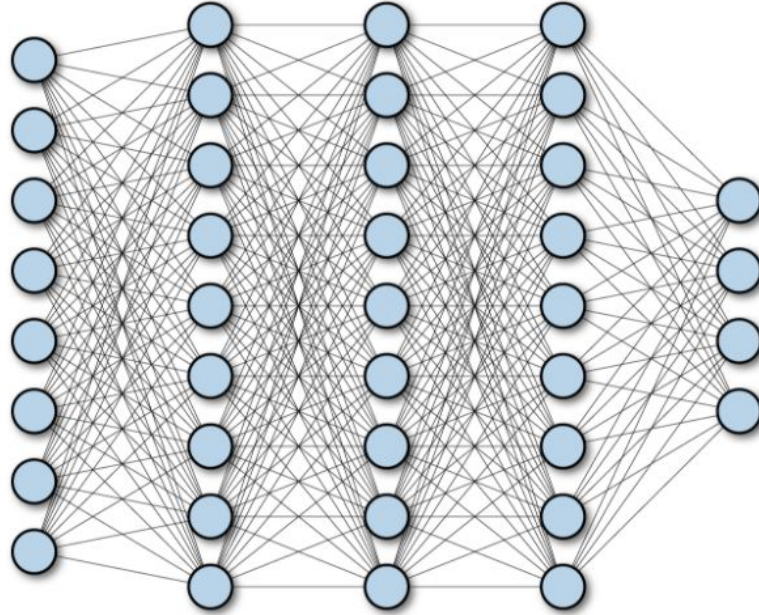
Inferência



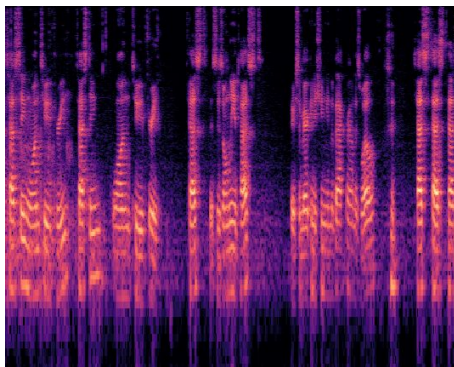
Programação com dados e aprendizado de máquinas



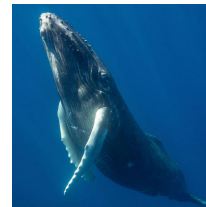
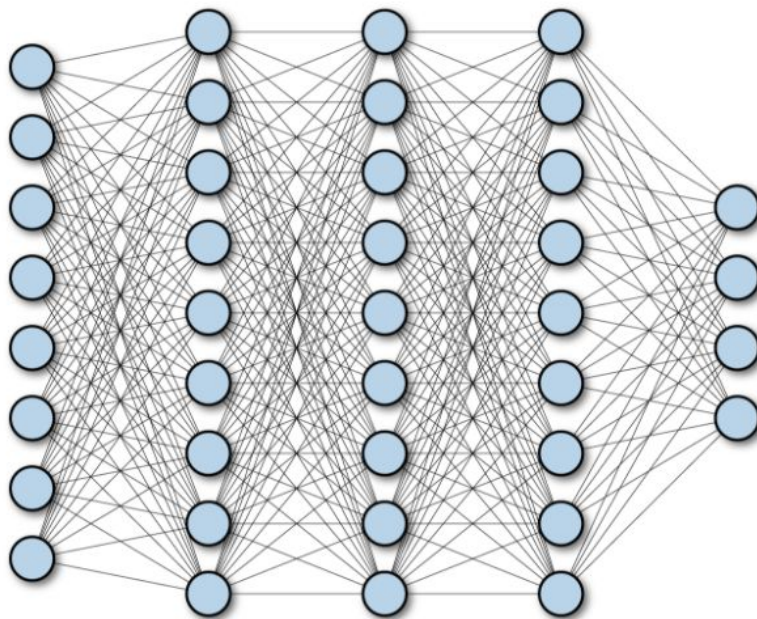
Programação com dados e aprendizado de máquinas



Programação com dados e aprendizado de máquinas



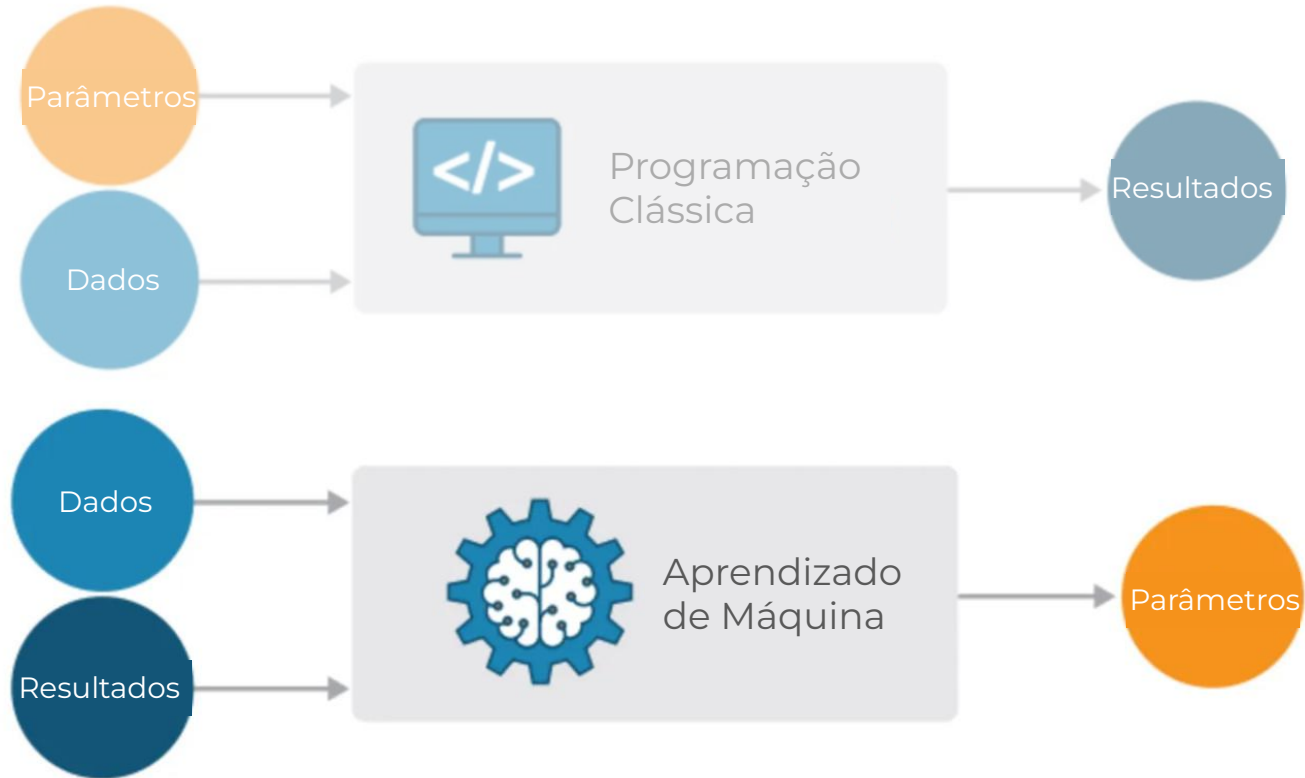
áudio: espectrogramas



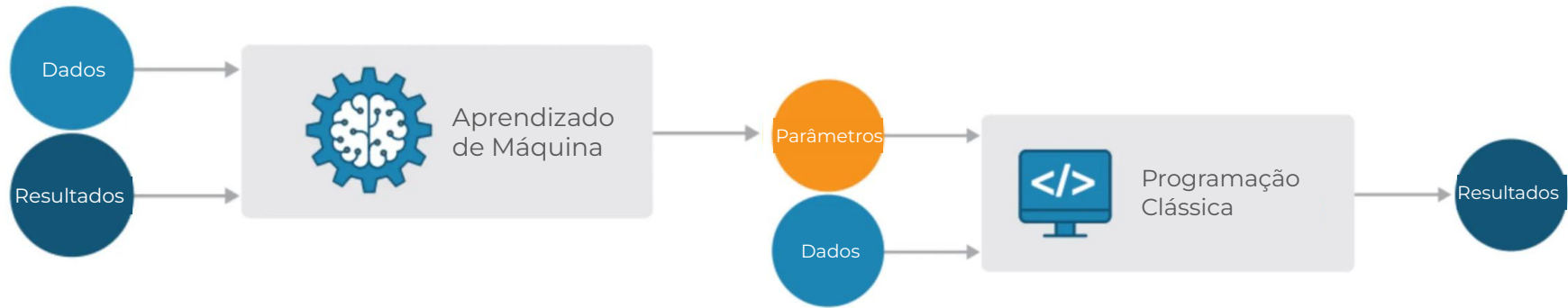
Paradigmas de programação



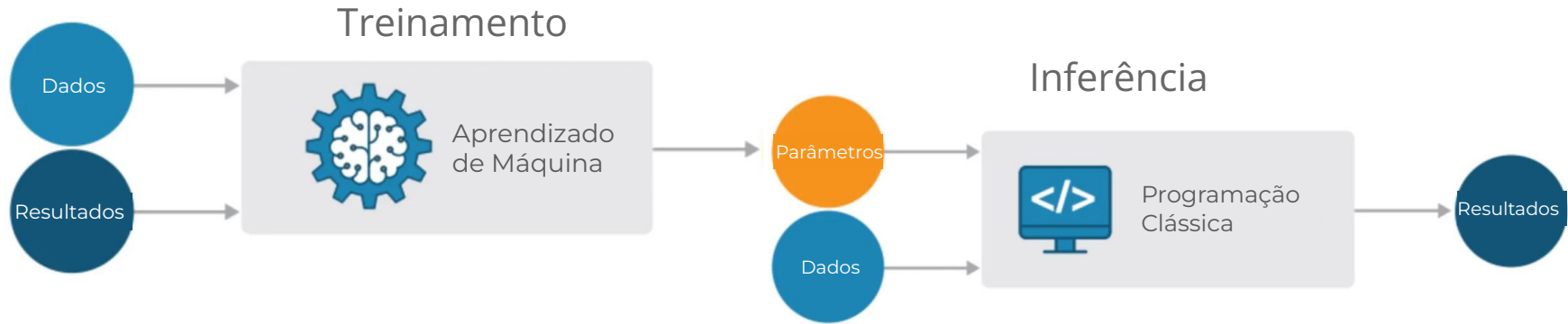
Paradigmas de programação



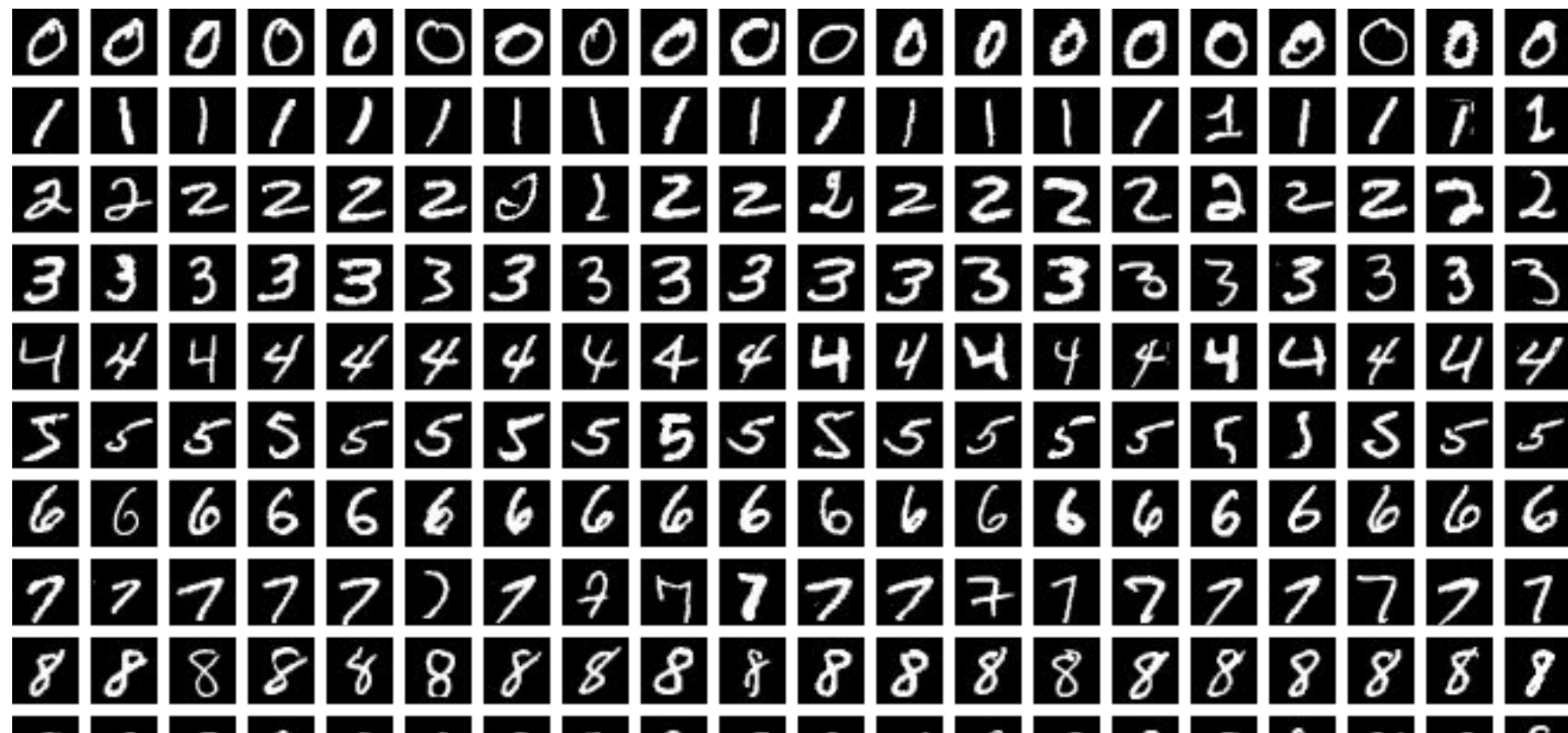
Paradigmas de programação



Paradigmas de programação



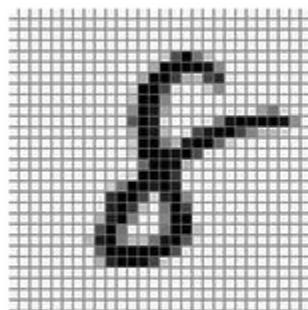
1990



1990



28



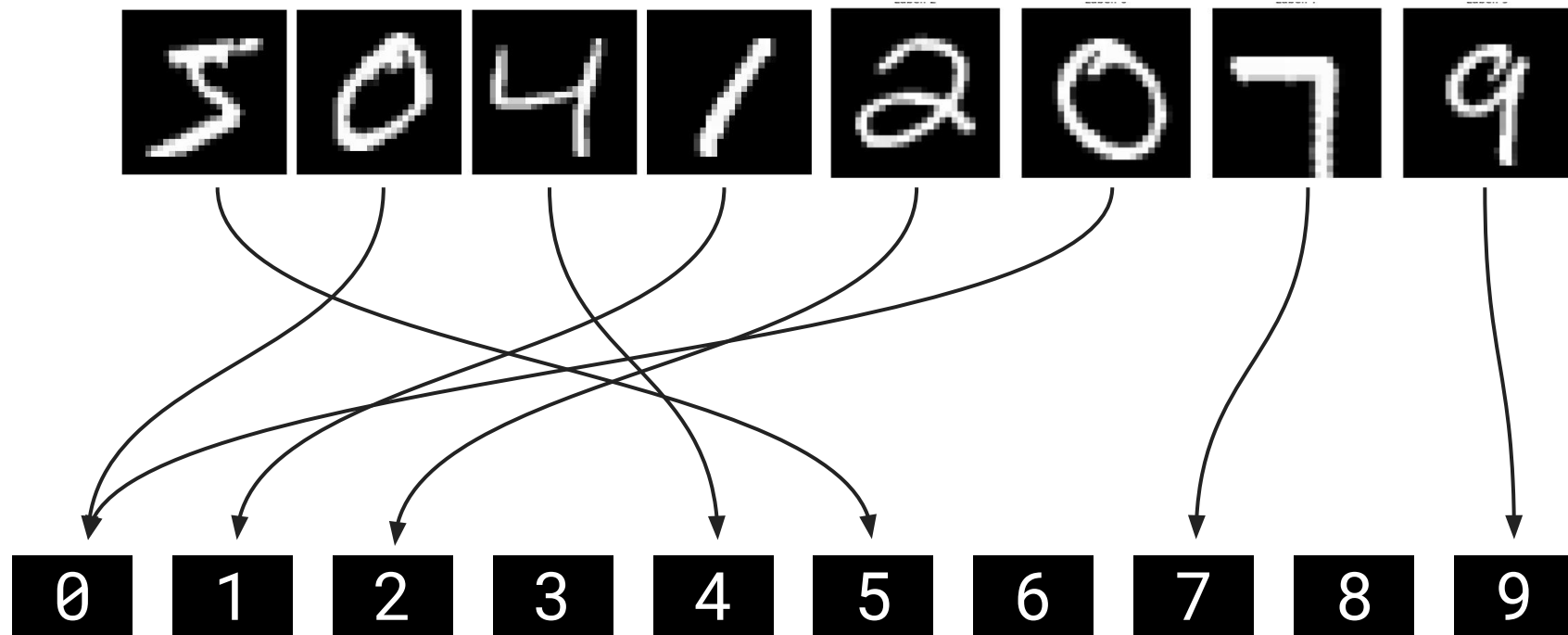
28

tarefas

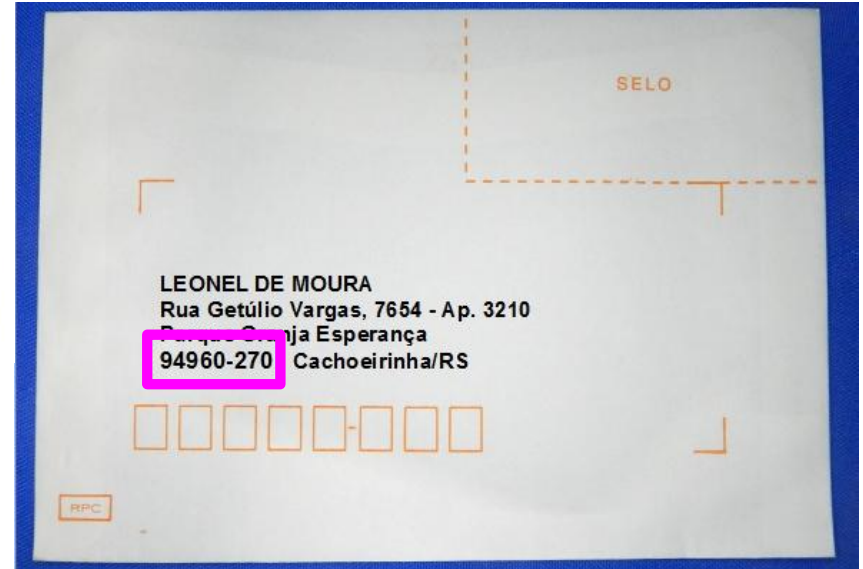
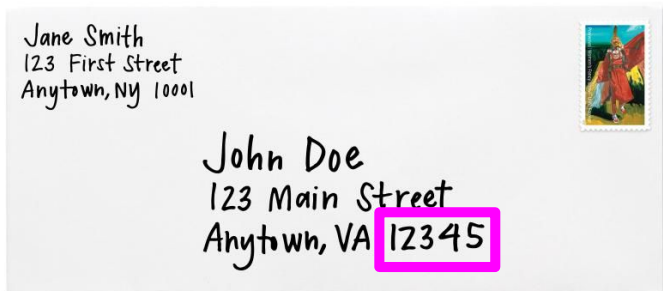
Tarefas: Classificação



Tarefas: Classificação



Tarefas: Regressão



Tipos de Modelos / Tarefas

- Classificação: rotular objetos
 - Sim / Não
 - Cachorro / Gato / Dinossauro
 - Qual letra / número
- Regressão: previsão de valores contínuos
 - Preço, temperatura, localização do CEP em envelopes

Modelos

 **Hugging Face**

Search models, datasets, users...

Models Datasets Spaces Community Docs Enterprise Pricing Log In

Tasks

Hugging Face is the home for all Machine Learning tasks. Here you can find what you need to get started with a task: demos, use cases, models, datasets, and more!

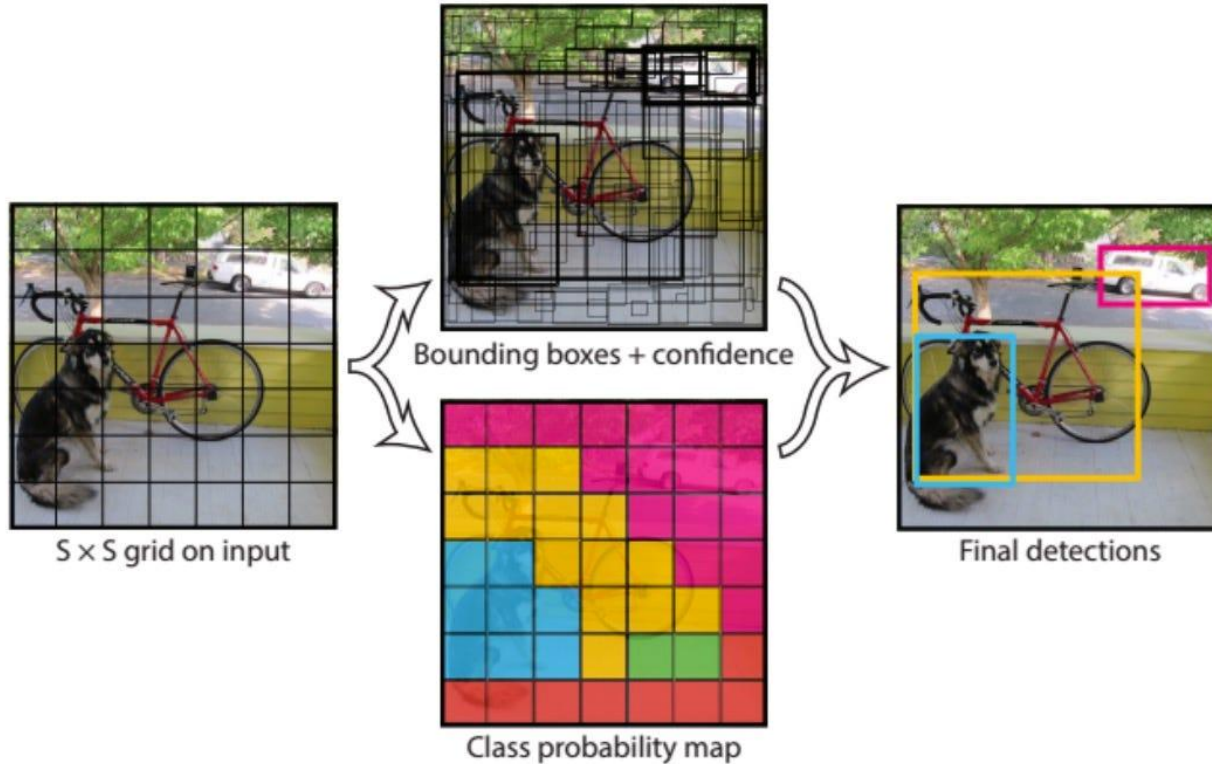
Multimodal

 Any-to-Any 11,408 models	 Audio-Text-to-Text 168 models	 Document Question Answering 240 models	 Visual Document Retrieval 129 models	 Image-Text-to-Text 20,026 models	 Image-Text-to-Image 41 models	 Image-Text-to-Video 12 models	 Video-Text-to-Text 272 models	 Visual Question Answering 619 models
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Natural Language Processing

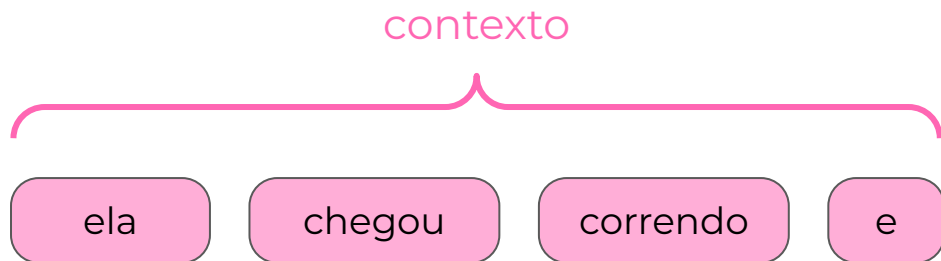
								
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Detecção de Objetos: Classificação + Regressão



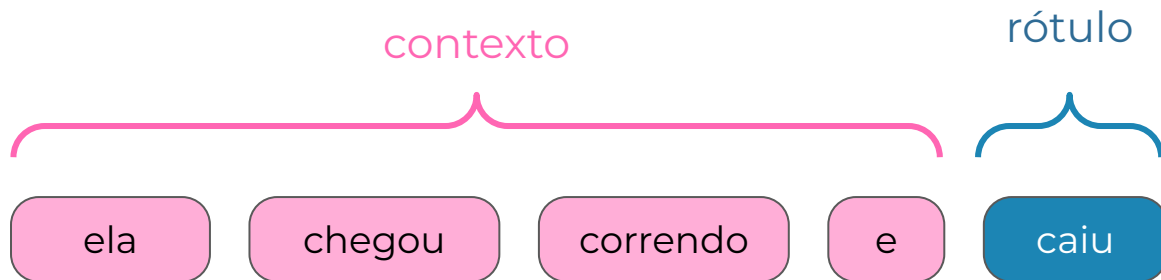
Geradores de texto (LLMs) ...

- ... são classificadores
 - que escolhem uma categoria dado um certo contexto



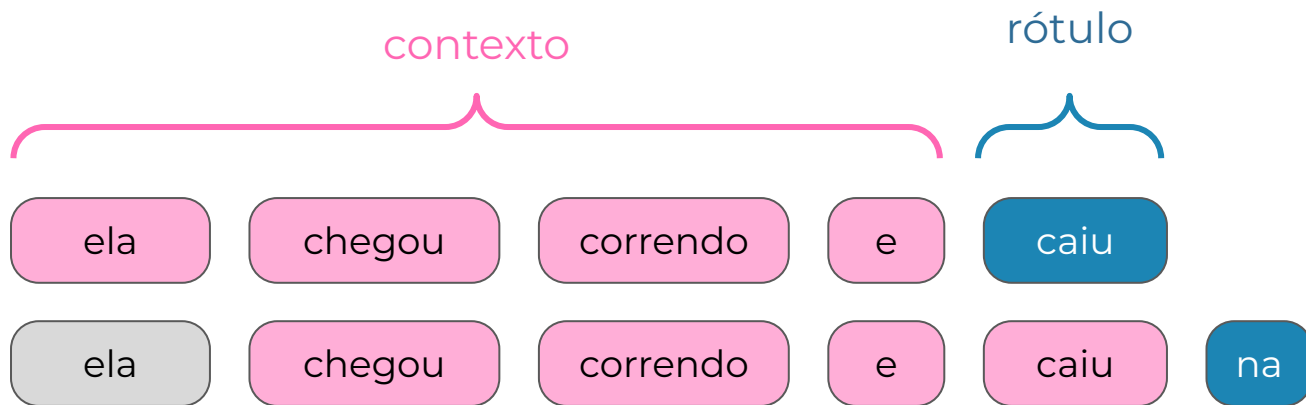
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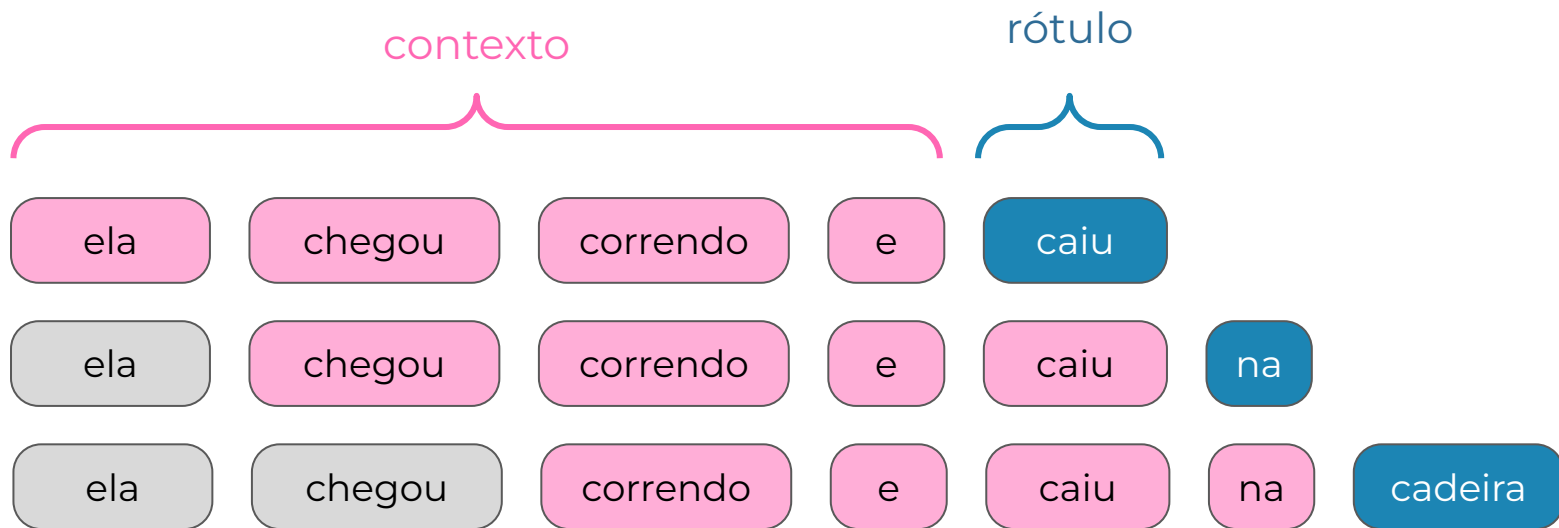
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Geradores de texto (LLMs) ...

- ... são classificadores
 - que escolhem uma categoria dado um certo contexto



Tipos de Modelos / Tarefas

- Classificação: agrupamento de objetos com rótulos pré-determinados
 - Sim / Não
 - Cachorro / Gato / Dinossauro
 - Qual letra / número
- Regressão: previsão de valores contínuos
 - Preço, temperatura, posicionamento do CEP no envelope

futuro

Próximos workshops

- Como usar modelos abertos
 - Com Python
 - Rodando no Google Collab ou Kaggle
- Como ajustar modelos de classificação
- Como extrair *embeddings* de textos e imagens

